MILITARY SPECIFICATION SHEET

FITTINGS, WIRE REINFORCED HYDRAULIC HOSE AND CLAMP-HALVES: FIELD ATTACHABLE (REUSABLE), SCREW-ON TO 37° FLARE FEMALE SWIVEL, 90° BENT TUBE, LONG DROP, TYPE 100R1, FOR USE WITH TYPE 100R1 SINGLE WIRE BRAID HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the fittings described herein shall consist of this specification and the latest issue of MIL-F-52525.

REQUIREMENTS:

1. The fitting shall conform to figure 1 and table I. The use of the alternate termination flare as shown in figure 2 and corresponding dimensions of table II is at the option of the contractor.

AMSC N/A FSC 4730
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TABLE I. Dimensions for type 100R1 fittings.1/

Specification part number	Hose dash number	Nominal hose I.D.	Tube dash number	Nominal tube O.D.	Nominal thread size	H Nominal hex size	L Maximum	
		inch		inch		inch	inch	mm
M52525/8-4-4	4	1/4	4	1/4	7/16-20	9/16	0.87	22.09
M52525/8-4-5	4	1/4	5	5/16	1/2-20	5/8	0.87	22.09
M52525/8-6-6	6	3/8	6	3/8	9/16-18	11/16	1.09	27.68
M52525/8-8-8	8	1/2	8	1/2	3/4-16	7/8	1.20	30.48
M52525/8-8-10	8	1/2	10	5/8	7/8-14	1	1.20	30.48
M52525/8-12-12	12	3/4	12	3/4	1-1/16-12	1-1/4	1.59	40.38
M52525/8-16-16	16	1	16	1	1-5/16-12	1-1/2	2.02	51.30

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TABLE I. Dimensions for type 100R1 fittings.1/ (Continued)

Specification part number	Hose dash number	Nominal hose I.D. inch	Tube dash number	Nominal tube O.D. inch	Nominal thread size	M Minimum inch	M Minimum mm	N inch +0.030 -0.000	N mm +0.76 -0.00
M52525/8-4-4	4	1/4	4	1/4	7/16-20	0.300	7.62	0.344	8.74
M52525/8-4-5	4	1/4	5	5/16	1/2-20	0.331	8.41	0.375	9.52
M52525/8-6-6	6	3/8	6	3/8	9/16-18	0.333	8.46	0.375	9.52
M52525/8-8-8	8	1/2	8	1/2	3/4-16	0.375	9.52	0.422	10.72
M52525/8-8-10	8	1/2	10	5/8	7/8-14	0.461	11.70	0.500	12.70
M52525/8-12-12	12	3/4	12	3/4	1-1/16-12	0.469	11.91	0.562	14.27
M52525/8-16-16	16	1	16	1	1-5/16-12	0.563	14.30	0.594	15.09

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TABLE I. Dimensions for type 100R1 fittings.[1/] (Continued)

Specification part number	Hose dash number	Nominal hose I.D. inch	Tube dash number	Nominal tube O.D. inch	Nominal thread size	P inch ±0.005	P mm ±0.13	Q Minimum inch	Q Minimum mm
M52525/8-4-4	4	1/4	4	1/4	7/16-20	0.295	7.49	0.11	2.79
M52525/8-4-5	4	1/4	5	5/16	1/2-20	0.355	9.02	0.11	2.79
M52525/8-6-6	6	3/8	6	3/8	9/16-18	0.435	11.04	0.26	6.60
M52525/8-8-8	8	1/2	8	1/2	3/4-16	0.570	14.48	0.38	9.65
M52525/8-8-10	8	1/2	10	5/8	7/8-14	0.680	17.27	0.38	9.65
M52525/8-12-12	12	3/4	12	3/4	1-1/16-12	0.850	21.59	0.57	14.48
M52525/8-16-16	16	1	16	1	1-5/16-12	1.000	25.40	0.78	19.81

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TABLE I. Dimensions for type 100R1 fittings.1/ (Continued)

Specification part number	Hose dash number	Nominal hose I.D.	Tube dash number	Nominal tube O.D.	Nominal thread size	C Maximum		F Maximum	
		inch		inch		inch	mm	inch	mm
M52525/8-4-4	4	1/4	4	1/4	7/16-20	2.63	66.80	1.86	47.24
M52525/8-4-5	4	1/4	5	5/16	1/2-20	2.67	67.82	1.86	47.24
M52525/8-6-6	6	3/8	6	3/8	9/16-18	3.03	76.96	2.24	56.90
M52525/8-8-8	8	1/2	8	1/2	3/4-16	3.56	90.42	2.49	63.25
M52525/8-8-10	8	1/2	10	5/8	7/8-14	3.78	96.01	2.63	66.80
M52525/8-12-12	12	3/4	12	3/4	1-1/16-12	4.28	108.71	3.79	96.27
M52525/8-16-16	16	1	16	1	1-5/16-12	5.20	132.08	4.39	111.51

1/ All dimensions not shown shall conform to SAE J516.

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TABLE II. Dimensions for alternate termination.1/

Tube size (REF)	D				M		N	
	Inch		Millimeter		Minimum		Minimum	
inch	max	min	max	min	inch	mm	inch	mm
1/4	0.360	0.340	9.14	8.64	0.295	7.5	0.375	9.52
5/16	0.430	0.400	10.92	10.16	0.295	7.5	0.375	9.52
3/8	0.490	0.460	12.44	11.69	0.297	7.5	0.385	9.78
1/2	0.660	0.630	16.76	16.01	0.339	8.6	0.438	11.13
5/8	0.790	0.760	20.06	19.31	0.409	10.4	0.521	13.23
3/4	0.950	0.920	24.13	23.37	0.464	11.8	0.594	15.09
1	1.200	1.170	30.48	29.72	0.495	12.6	0.625	15.88

1/ Dimensions H, Q, C, and F shall be in accordance with table I.

2. The maximum operating pressures shall be as specified in table III.

TABLE III. Maximum operating pressures of fittings.

Hose I.D. (REF) (inch)	PSI (pounds per square inch)	MPa (megapascals)
1/4	2750	19.0
5/16	2500	17.2
3/8	2250	15.5
1/2	2000	13.8
5/8	1500	10.3
3/4	1250	8.6
1	1000	6.9

3. The over-tightening torque for swivel nuts shall be as specified in table IV.

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TABLE IV. Over-tightening torque for swivel nut.

Nominal tube O.D. (REF) (inch)	Torque	
	inch pounds	Newton meter (Nm)
1/4	135	15.3
5/16	180	20.3
3/8	270	30.5
1/2	450	50.8
5/8	650	73.4
3/4	900	101.7
1	1200	135.6

1/ Torque is expressed as Newton meter per radian (N·m/rad)·rad = N·m.

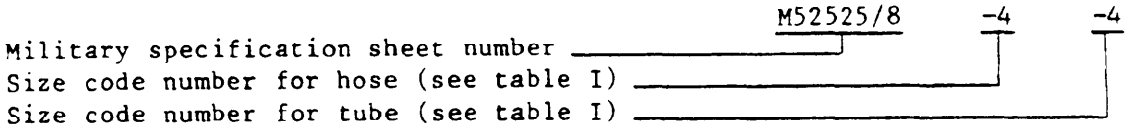
4. Marking shall include the manufacturer's name or trademark, size, and hose identifier R-1. Location of marking is optional.

5. This item must be listed on QPL-52525.

INTENDED USE: Type 100R1 fittings are for use with single wire braid reinforced hose conforming to MIL-H-52471, Type 100R1.

CATALOGING DATA: The military specification part number is a definitive part number which corresponds to the size fitting covered by this specification sheet. The military specification number, the military specification sheet number, code number for size of hose, and code number for size of tube are combined to form the military specification part number.

EXAMPLE: Military specification part number



The approved item name of this item, in accordance with the Federal Item Name Directory for the Federal Catalog System, Handbook H6, Section A, is Elbow, Tube to Hose.

Custodian:
Army - ME

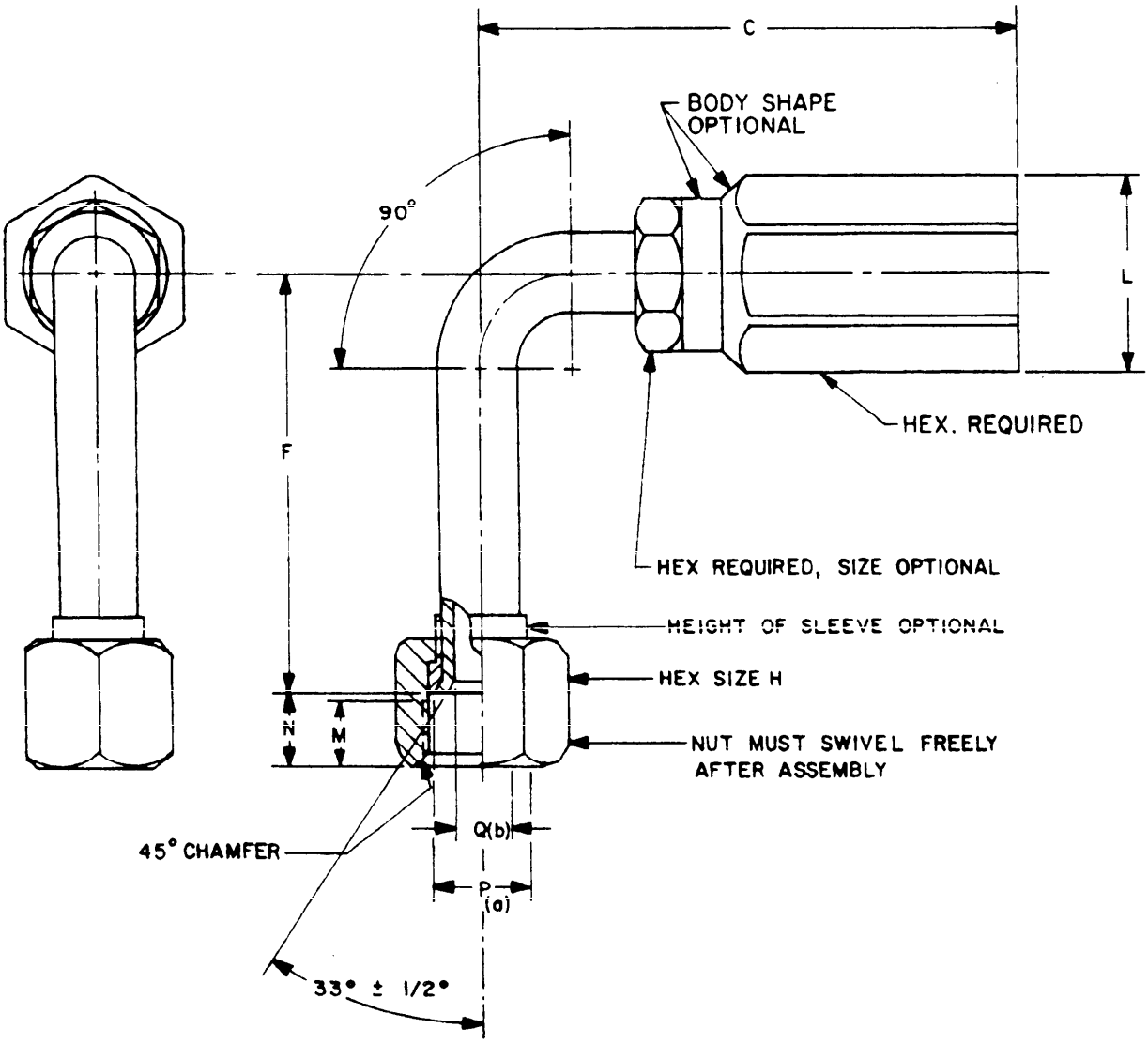
Preparing activity:
Army - ME

Review activities:
Army - GL
DLA - CS

Project 4730-0524

User activity:
Air Force - 99

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(a) DIMENSION P TO BE CONCENTRIC WITH DIMENSION Q WITHIN 0.015 FIM.
(b) Q = BORE DIAMETER

FIGURE 1. Fitting screw-on to 37° flare, 90° bent tube long drop.

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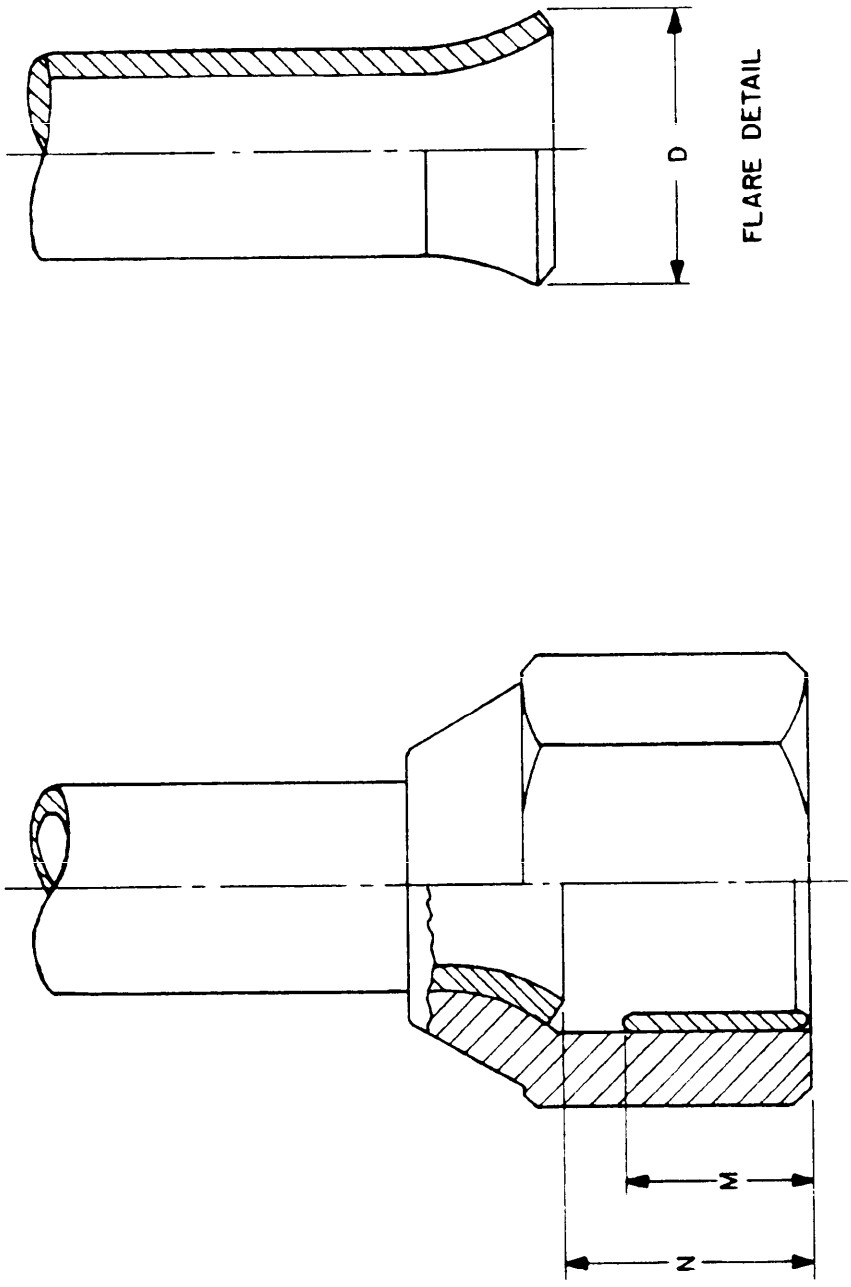


FIGURE 2. Alternate termination radial flare.

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